Vineeth Raghav Gururajan

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PROFILE

M.Sc. Electrical Engineering student at the University of Stuttgart (power systems, power electronics, battery modeling) with a hands-on B.Tech. background in embedded systems, FPGA design in Verilog and machine learning. Seeking a working student position in electrical development, hardware, simulation or power systems. Available up to 20 hours per week.

EDUCATION

M.Sc. Electrical Engineering – University of Stuttgart (10/2025 – expected 2027)

- Intelligent Power Grids, Power Systems Engineering, Battery Modeling

B.Tech. Electronics and Computer Engineering – SRM Institute of Science and Technology, Chennai (2021 – 2025), CGPA 8.70/10

- Grades A+ or A in FPGA-Based Embedded Systems, Embedded Hardware and Operating Systems, Microcontrollers and Interfacing, Hardware Interfacing and Networking, IoT System Design, Deep Learning; final-year project graded A+ (10 credits)
- Seminar I (2023, grade O): cryptocurrency price analysis and prediction, review of forecasting methods (technical indicators, linear regression, random forest, LSTM) and evaluation metrics (MAE, RMSE)

PROJECTS

Battery Cell Modeling – team project, University of Stuttgart (2026)

- Analysed cell-to-cell variance caused by electrical, thermal, aging and manufacturing tolerances by simulation; implemented and debugged statistical parameter sampling in the simulation model

Optimized 64-bit RoBA Multiplier with MAC Unit – B.Tech. final-year project, Verilog, Xilinx ISE (2025)

- Designed a rounding-based approximate multiplier with a multiply-accumulate unit that replaces partial-product computation with shift-and-add operations; synthesised in Xilinx ISE (Spartan-3 target) with a reported critical path of 0.575 ns

Duplicate Question Detection – team project (4 students), Python, machine learning (2024)

- Classified duplicate question pairs from the Quora Question Pairs dataset using NLP and fuzzy-ratio features; SVM reached 88.3% and Random Forest 80.02% accuracy; deployed as a web application on Heroku

SKILLS

Electronics and embedded: microcontrollers (ARM, ESP32, Arduino), FPGA design (Verilog, Xilinx ISE), PCB design (Altium), circuit simulation (PSpice), communication protocols (CAN, I2C, SPI, UART)

Software and modeling: Python, C/C++, MATLAB, machine learning, statistical parameter sampling

Power and control: power electronics, control systems, sensors and actuators, battery modeling

LANGUAGES

English: C1 (IELTS Academic, overall band 7.5, October 2024) · **German:** A2, preparing for the telc B1 exam

CERTIFICATES AND VOLUNTEERING

PSpice Fundamentals (TVS India); Certificate of Training (Livewire Ltd.) · STEM workshops in electronics and coding for underprivileged students; post-cyclone relief work in Chennai